

11 Answer: A

$$\phi = -\frac{GM}{r}$$

$$r_A = -\frac{GM}{\phi_A} \quad \text{and} \quad r_B = -\frac{GM}{\phi_B}$$

$$BA = r_B - r_A$$

$$= -GM \left(\frac{1}{\phi_B} - \frac{1}{\phi_A} \right)$$

$$= 0.89 \times 10^6 \text{ m}$$

$$r_C = -\frac{GM}{\phi_C} \quad \text{and} \quad r_B = -\frac{GM}{\phi_B}$$

$$CB = r_C - r_B$$

$$= -GM \left(\frac{1}{\phi_C} - \frac{1}{\phi_B} \right)$$

$$= 1.11 \times 10^6 \text{ m}$$